



BITS Pilani presentation

BITS Pilani
Pilani Campus

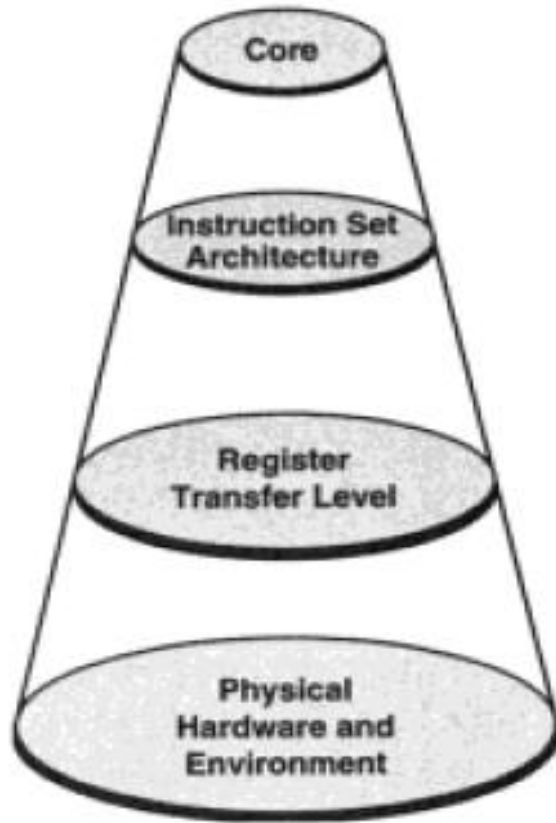
Faculty Name
Faculty Department



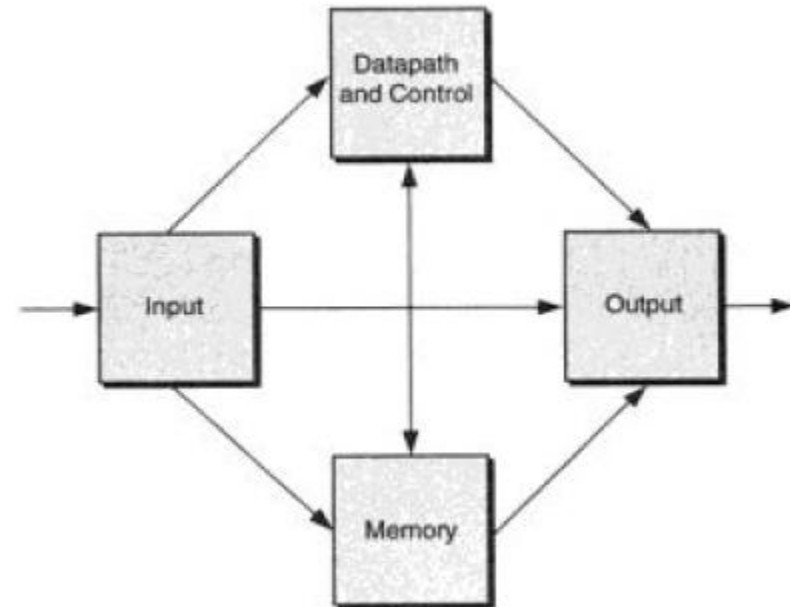
EEE ZG512 / SS ZG512, EMBEDDED SYSTEM DESIGN

Lecture No. 03

Hardware Components:



Core level

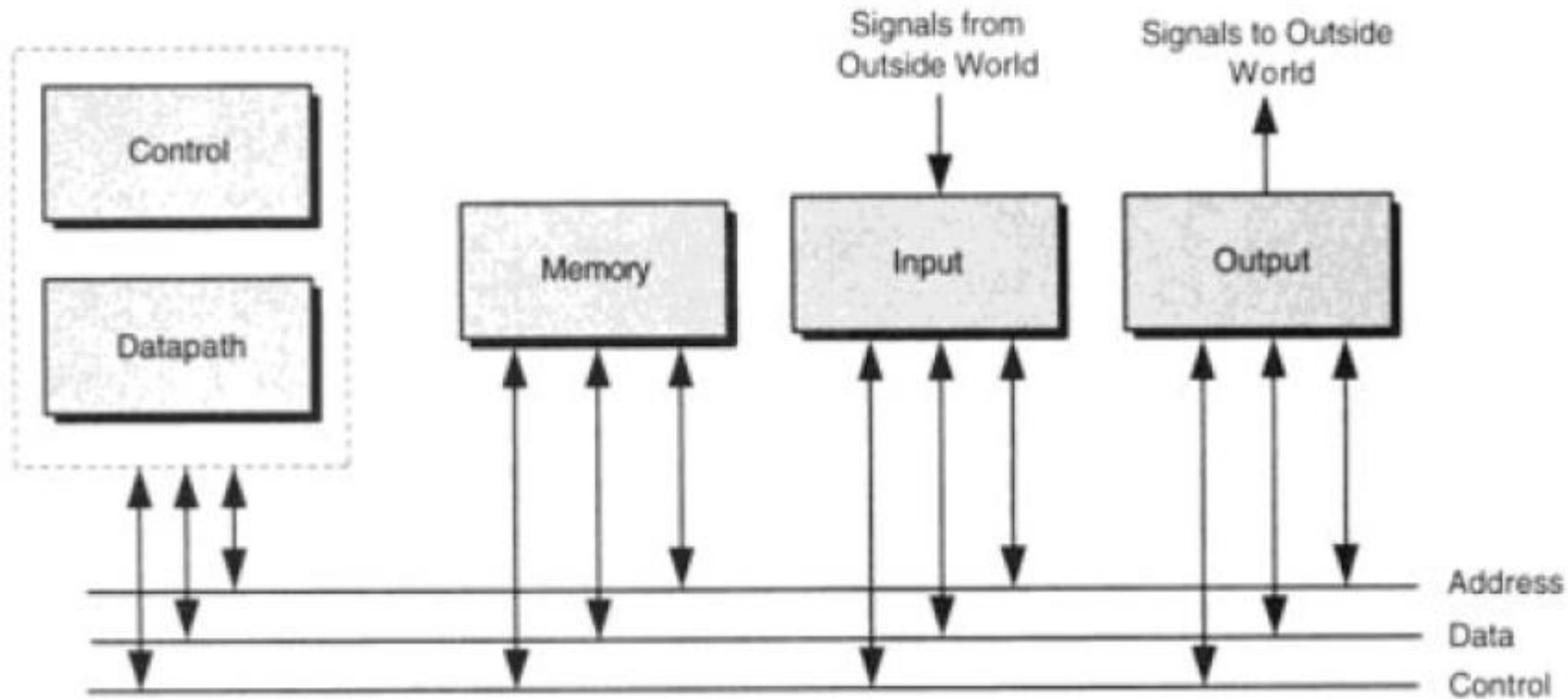


Bus Structure

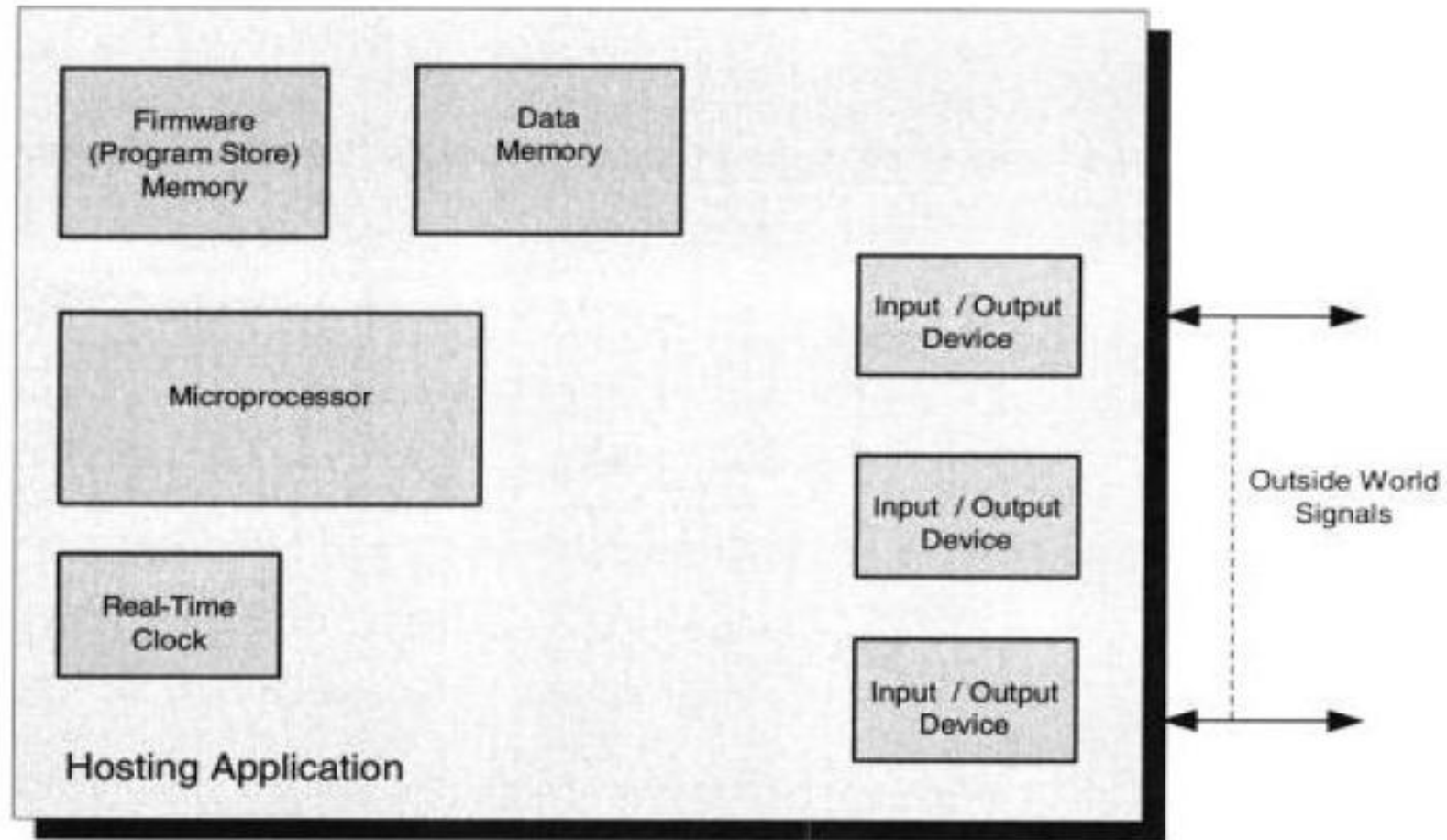
innovate

achieve

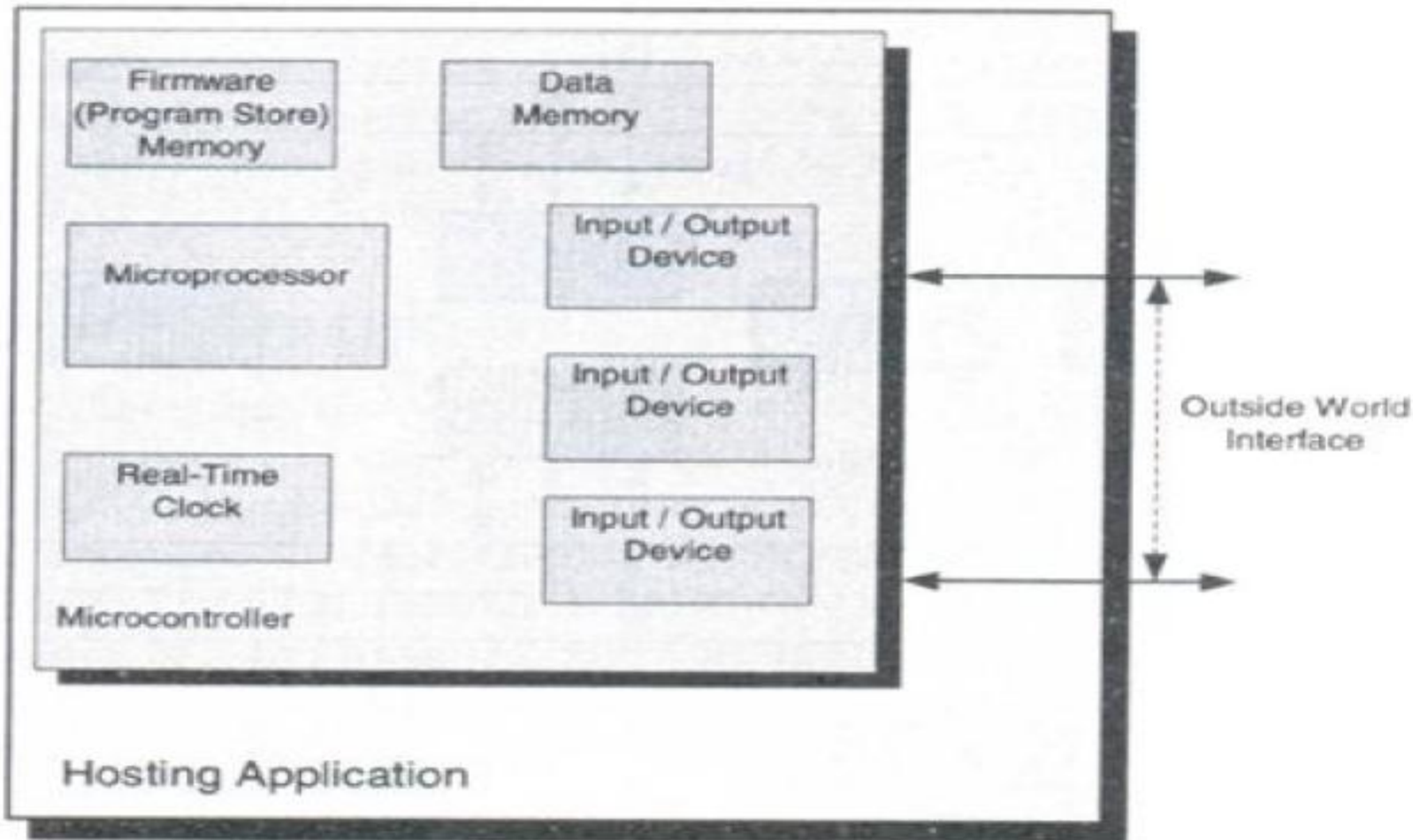
lead



Microprocessor based system



Microcontroller bases system

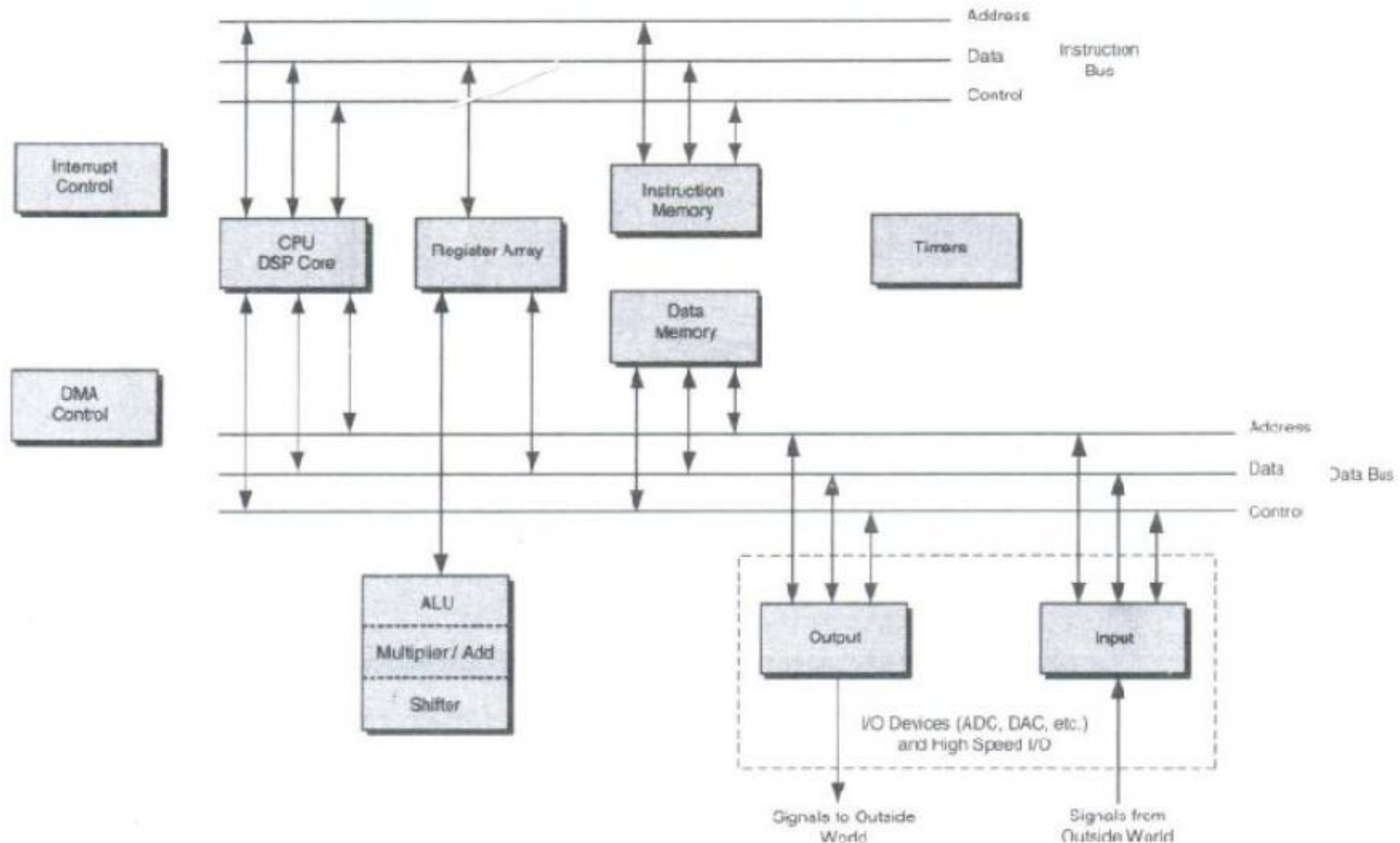


Digital Signal Processor

innovate

achieve

lead



Important features



- Hardware modulo addressing.
- Extensive usage of DMA (Direct Memory Access).
- Harvard Architecture is preferred.
- SIMD (Single Instruction Multiple Data) instructions.
- VLIW (Very Long Instruction Word).
- MAC unit (Multiply and accumulate).

Information storage (Endianness)

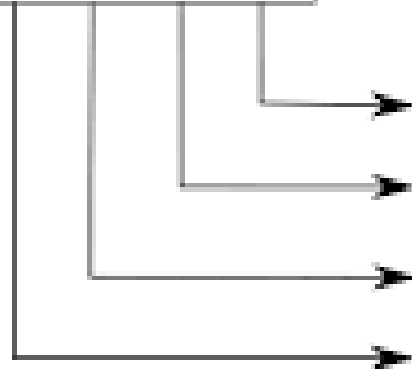


- Little Endian
- Big Endian

Little-endian

32-bit integer

0A0B0C0D



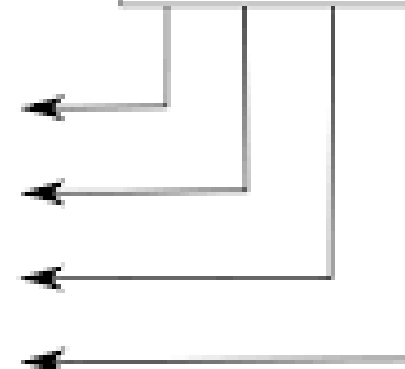
Memory

⋮		⋮
0D	a	0A
0C	$a+1$	0B
0B	$a+2$	0C
0A	$a+3$	0D
⋮		⋮

Big-endian

32-bit integer

0A0B0C0D



Addressing Modes

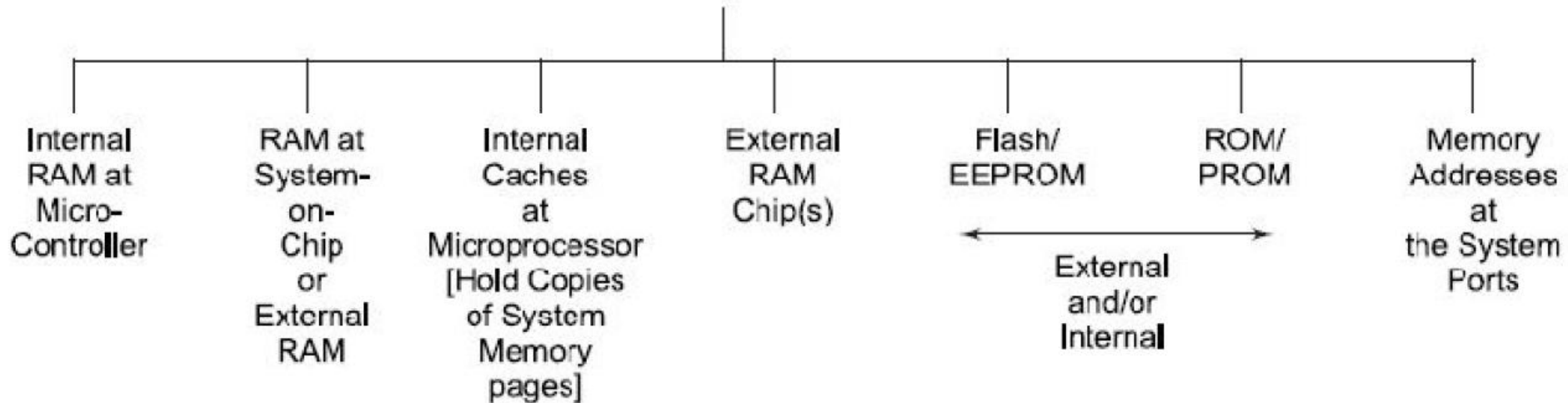


- Immediate: Data in instruction – Ex: MOV R0, #15
- Register: Data in GPR – Ex: MOV R0, R1
- Direct: Address of data - Ex: LDR R0, Address
- Register Indirect: GPR contain address of data –
Ex: LDR R2, [R0]
- *Implicit: No operand - Ex: RRC, RLC*
- Indexed: $Base + offset = 0x10000000 + 0xFF = 0x100000FF$
- $MOV C A, A + DPTR, MOV C A, A + PC$

Different forms of System Memory



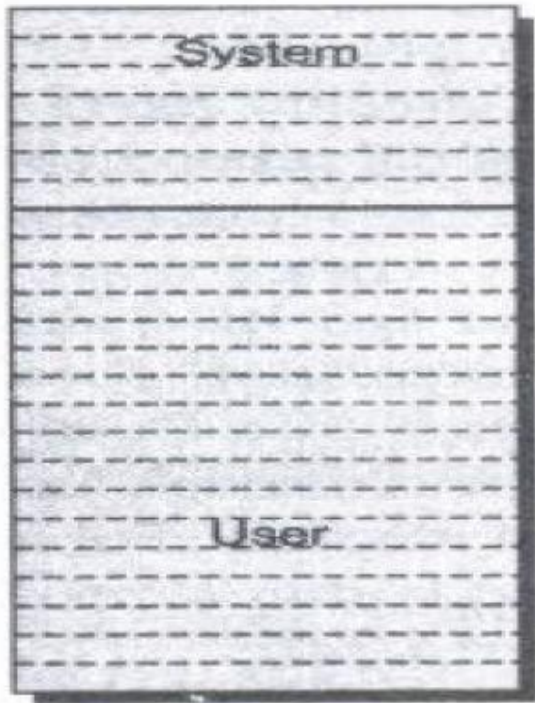
Various Forms of System Memory



Memory Hierarchy



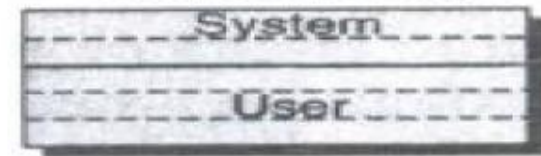
Secondary Memory



Main Memory



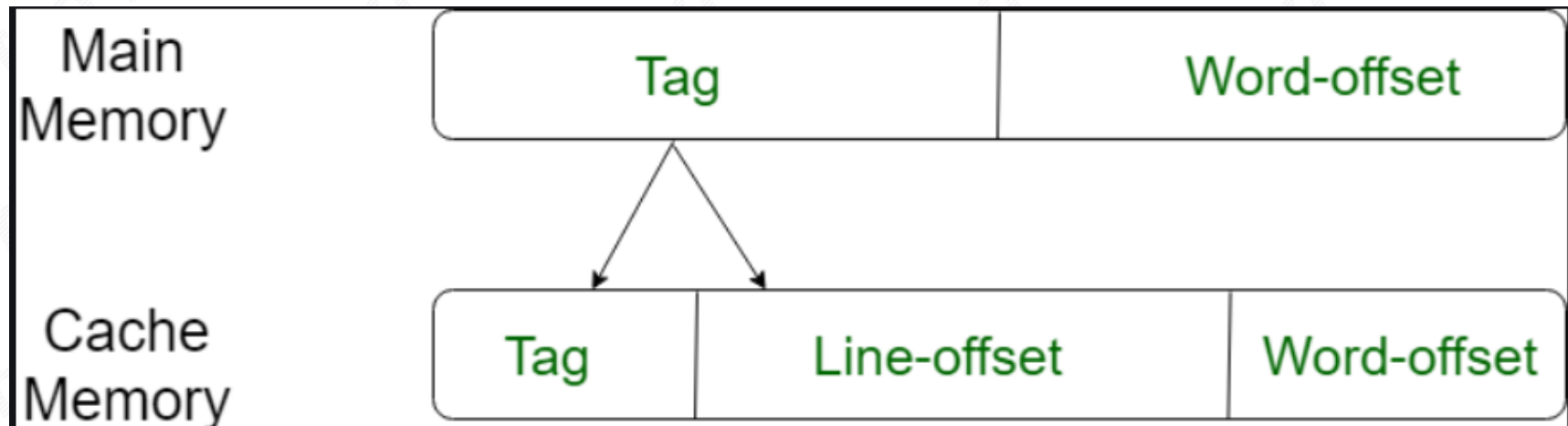
Cache Memory



Cache Mapping Techniques



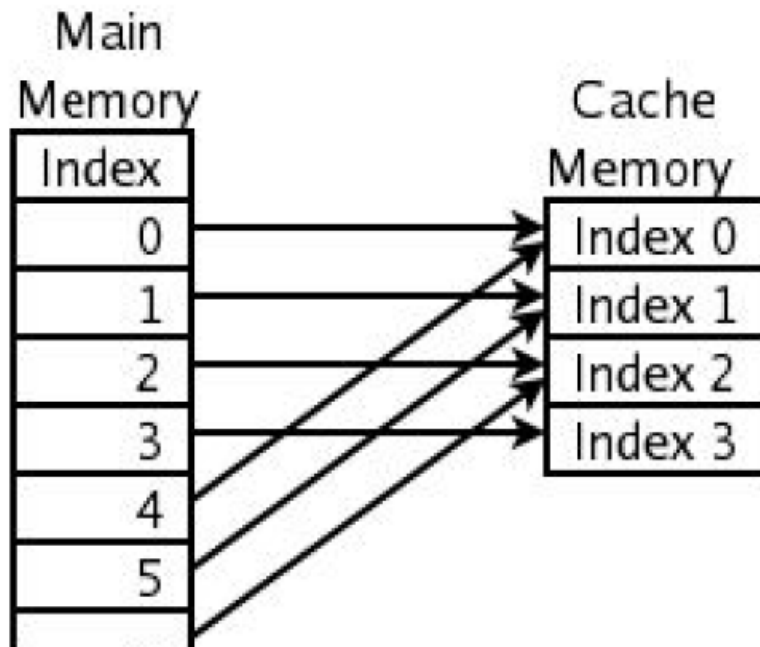
- Direct mapping:
- Associative mapping
- Set Associative mapping



Cache Mapping

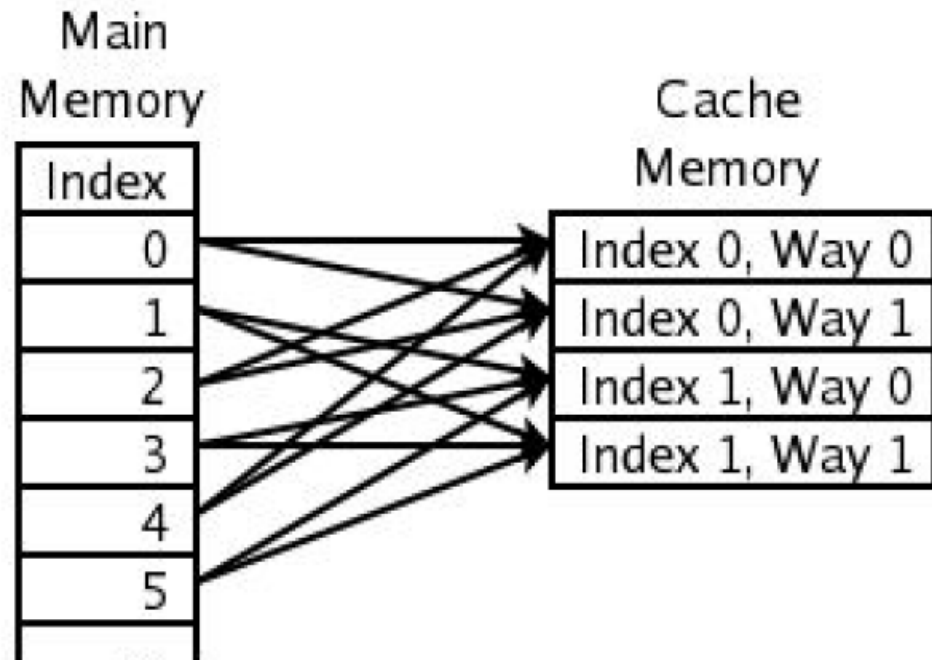


Direct Mapped
Cache Fill



Each location in main memory can be cached by just one cache location.

2-Way Associative
Cache Fill



Each location in main memory can be cached by one of two cache locations.

END
of
Session – 3
Thank You

